Phase 5: Implementation and Evaluation

Overview

In this phase, you will implement the interpretable deep learning method selected in Phase 4 and apply it to the problem identified in Phase 3. You will adapt the model architecture based on the challenges and computational constraints you anticipated in Phase 4, then conduct an experimental analysis of its performance and interpretability.

You will:

- Implement a computationally feasible version of your chosen interpretable method, incorporating the modifications proposed in Phase 4.
- Adapt the architecture to address domain-specific challenges identified in previous phases.
- Conduct systematic experiments to evaluate performance across multiple metrics.
- Analyze the interpretability in the medical context.
- Compare results against the validation protocol established in your Phase 3 paper.
- Document trade-offs between accuracy, interpretability, computational cost, and clinical utility.

This phase emphasizes practical implementation skills, experimental rigor, and the ability to translate research-grade models to real-world medical applications.

Step 1: Implementation and Adaptation (40%)

1. Implement the exact validation protocol used in your Phase 3 paper. (Follow the same data split, and evaluation metrics.)
2. Implement a computationally feasible version of your selected interpretable method (from Phase 4) that incorporates the optimizations you proposed in Phase 4.
3. Further modify the architecture to address domain-specific challenges identified in Phases 3 and 4. You may conduct some tests to adapt the hyperparameters of the model and the loss function to account for your specific experimental setup.
4. Document the specific compromises made compared to the original paper's implementation (from phase 4) due to resource constraints.

⚠️ When conducting your hyperparameter study, start with the original hyperparameter values reported in the paper of your chosen method (from Phase 4). However, due to computational constraints (e.g., limited GPU memory), you may need to reduce the batch size—this is acceptable and expected. Also, remember to report the values of all other hyperparameters.

Step 2: Experimentation and Analysis (60%)

1. Conduct a systematic study of at least three critical hyperparameters:
    - Define a range of values\choices for each hyperparameter
    - Analyze the impact of each hyperparameter on both performance metrics, interpretability quality, and computational cost trade-offs.
    - Visualize hyperparameter effects using appropriate plots.
2. Compare the results against the original results from your Phase 3 paper across performance metrics (e.g., accuracy, precision, inference time).
3. Analyze the practical clinical value of your model's interpretations. Do prototypes correspond to clinically meaningful features? Plot some examples and answer the question based on those.

⚠️ For each of the hyperparameters you analyze, test at least three meaningful values/choices around the original setting (e.g., low, medium, high) to assess sensitivity. For example:

Number of prototypes: 10, 20, 30 per class

Encoder: 'resnet34', 'vgg19', 'densenet121'

Number of attention heads: 4, 8, 12

Avoid extreme values that lead to obvious instability or failure. This structured approach enables valid comparisons and helps identify optimal configurations.

Submission Instruction

The students should upload their report/notebook there according to this structure on their repository in https://github.com/UT-AI-Lab. The name of the submission file for this phase should be *"<Student Id>_Phase5"*.

├── *student_name_1/*

│ ├── *"<Student Id>_Phase5".ipynb # Main Jupyter notebook (answers + results)*

│ ...

⚠️ **Deadline for submission: 28 Azar.**